

Fluorination Reactions

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3,3-Difluoroallyl Sulfonium Salts: Practical and Bench-Stable Reagents for Highly Regioselective *gem*-Difluoroalkylations

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Abstract: The site-selective introduction of the difluoromethylene group into organic molecules has important applications in producing pharmaceuticals and agrochemicals. However, the general and efficient methods that can construct both C(sp²)-CF₂R and C(sp³)-CF₂R bonds remain challenging. Here, we disclose a new type of practical and bench-stable difluoroalkylating reagent 3,3-difluoroallyl sulfonium salt (DFAS) that can be practically prepared from inexpensive and bulk chemical feedstock 3,3,3-trifluoropropene. This reagent allows highly regioselective *gem*-difluoroalkylation of various organozinc reagents, including aryl, primary, secondary, and tertiary alkyl zinc reagents, via copper catalysis under mild reaction conditions with high efficiency. The reaction can also be extended to a series of substituted DFASs. Application of the approach leads to the short synthesis of complex analogs, showing the prospect of DFASs in medicinal chemistry.

Introduction

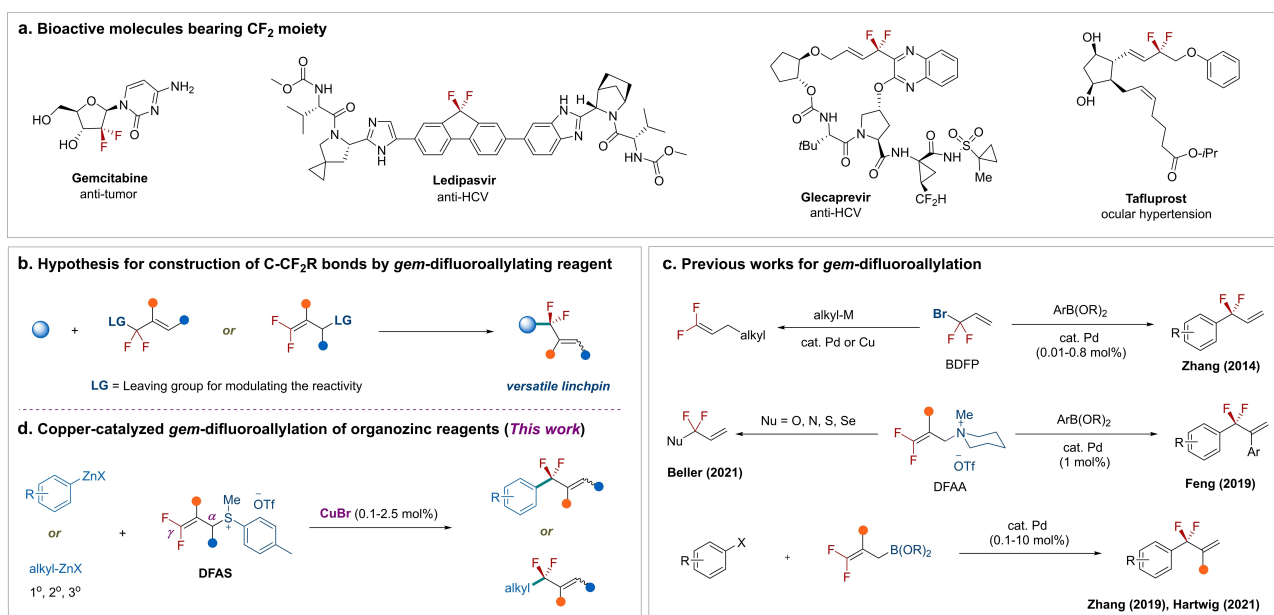
Due to the unique properties of fluorine atom(s),^[1] the tactical replacement of C–H bond(s) with C–F bond(s) has emerged as one of the crucial strategies in discovering new pharmaceuticals, agrochemicals, and advanced functional materials.^[2] Over the past decade, considerable efforts have been made to develop efficient methods for organofluorine compounds,^[3] and numerous fluorinating and fluoroalkylating reagents^[4] have been reported. However, most of the developed methods focus on the fluorination and trifluoromethylation reactions,^[3,4] the site-selective introduction of the difluoromethylene (CF₂) group into organic molecules remains less explored.^[3e,5] In particular, the development of a general and practical method that not only can construct C(sp²)-CF₂R bonds but also can forge C-

(sp³)-CF₂R bonds, including primary, secondary, and tertiary alkyl-CF₂R bonds, remains a challenging synthetic topic so far.

CF₂ group has been considered a bioisostere of oxygen atom^[1c,6] and polar functionalities, such as carbonyl.^[1c,6a] The precise introduction of CF₂ into biologically active molecules can significantly improve their metabolic stabilities, change the molecular conformations, and increase the acidity of the group neighboring CF₂ (Scheme 1a).^[1,7] As such, it is of great interest to develop general, practical, and efficient difluoroalkylation reactions for medicinal chemistry. We envisioned that developing a versatile and practical *gem*-difluoroalkylating reagent may provide an attractive opportunity to overcome the limitations in constructing C–CF₂R bonds (Scheme 1b). Our hypothesis is based on the consideration that 1) the *gem*-difluoroallyl group is a versatile synthetic handle that can be used for the diversified synthesis of difluoroalkylated compounds through transformations of its carbon-carbon double bond; 2) the installation of a suitable leaving group on the *gem*-difluoroalkylating reagent can fine-tune its reactivity, thereby, making it a general reagent to construct both C(sp²)-CF₂R and C-(sp³)-CF₂R bonds; 3) *gem*-difluoroallyl moiety has been found in several drugs for the treatment of hepatitis C virus (HCV)^[8] and ocular hypertension^[9] (Scheme 1a), the development of a practical and general *gem*-difluoroalkylating reagent and its applications in efficient *gem*-difluoroalkylation would provide a new tool in discovering interesting new bioactive molecules.

Traditionally, 3-bromo-3,3-difluoropropene (BDFP)^[10] has been used for *gem*-difluoroalkylation of aldehydes and imines.^[11] Recently, we reported a palladium-catalyzed highly regioselective *gem*-difluoroalkylation of arylborons with BDFP,^[12] providing efficient access to *gem*-difluoroalkylated arenes (Scheme 1c). However, the coupling of BDFP with aliphatic nucleophiles remains an unsolved problem due to the preferable formation of regioisomer *gem*-difluoroalkenes in the reaction (Scheme 1c).^[13] Very recently, two new types of *gem*-difluoroalkylating reagents have been reported, namely, 3,3-difluoroallyl ammonium salts (DFAAs)^[14] and *gem*-difluoroallylborons.^[15] Nevertheless, these two types of reagents are limited to the palladium-catalyzed formation of Ar–CF₂R bonds^[14a,15] or the metal-free construction of RCF₂–X (X = O, N, S, Se) bonds^[14b] (Scheme 1c). To date, the efficient construction of C(sp³)-CF₂R bonds with high selectivity and generality remains challenging,^[16] and no example of catalytic *gem*-difluoroalkylation of aliphatic substrates has been reported yet.

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Scheme 1. Strategies for *gem*-difluoroallylation and representative biologically active molecules bearing CF₂ moiety.

Here, we disclose a new type of bench-stable and practical *gem*-difluoroallylating reagent 3,3-difluoroallyl sulfonium salt (DFAS), prepared from inexpensive and bulk chemical feedstock 3,3,3-trifluoropropene (TFP). The application of DFAS to the copper-catalyzed cross-coupling leads to highly regioselective *gem*-difluoroallylation of a wide range of aromatic and aliphatic zinc reagents with low catalyst loading (0.1–2.5 mol%), providing a general and practical approach to construct both C(sp²)-CF₂R and C(sp³)-CF₂R bonds (Scheme 1d). Notably, primary, secondary, and tertiary alkyl zinc reagents are all applicable to the reaction with high efficiency and good functional group tolerance. The reaction can also be extended to various substituted 3,3-difluoroallyl sulfonium salts (DFASs), providing an array of versatile *gem*-difluoroallylated structures. The synthetic utility of this protocol has also been demonstrated by the rapid modification of bioactive molecules, providing a facile route for applications in medicinal chemistry.

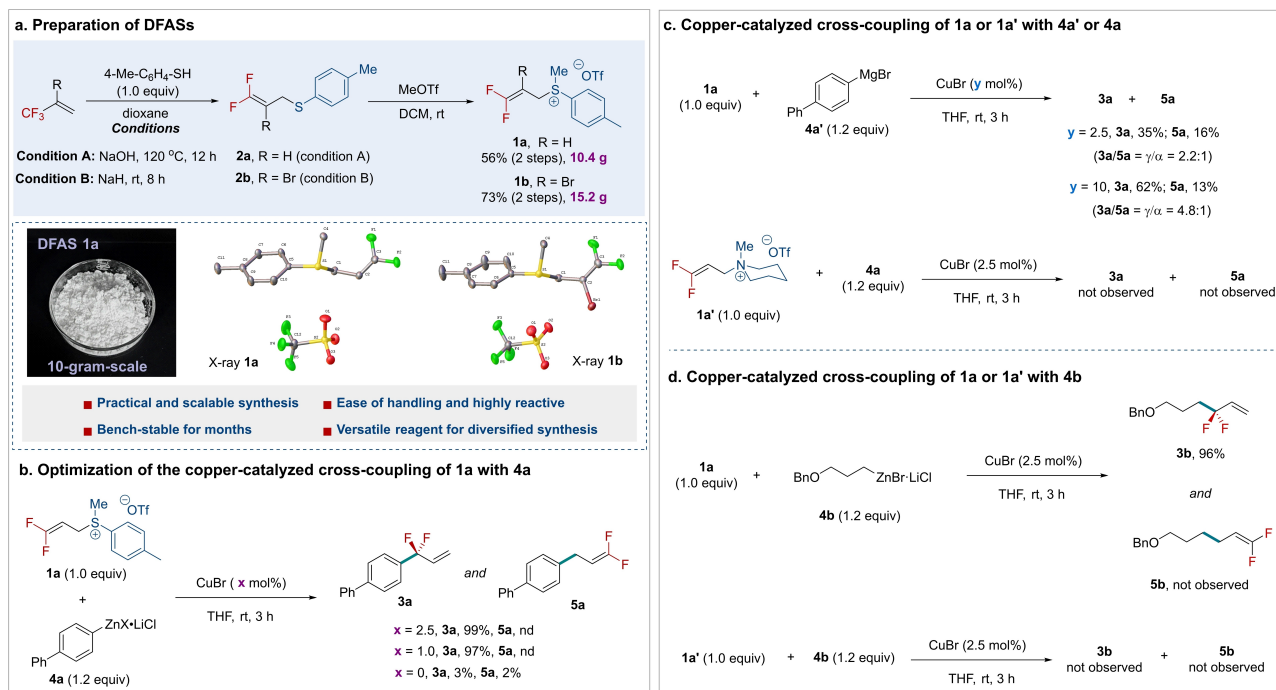
Results and Discussion

Preparation and Reactivity of DFASs

Sulphonium salts are useful synthetic intermediates in the construction of complex molecules.^[17] In particular, because of the strong electron-withdrawing nature of sulphonium salts, several electrophilic fluoroalkylating reagents bearing such a structural motif, such as the Umemoto reagent,^[18a] have been developed.^[18] However, the preparation of DFASs and their reactivity investigation have not been reported so far, although they represent a new type of electrophilic fluoroalkylating reagent. The sulphonium salts are more reactive than the ammonium salts to serve as a

leaving group. We envisioned that installing a sulphonium salt moiety at the γ -position of 1,1-difluoropropene would lead to a more reactive *gem*-difluoroallylating reagent because the preferential cleavage of its labile C–S bond would provide a strong driving force to construct the C–CF₂R bonds via an S_N2' pathway. Accordingly, we chose inexpensive and bulk industrial chemical 3,3,3-trifluoropropene (TFP) and low-cost and commercially available 4-methylbenzenethiol as the starting materials to prepare DFAS **1a** (Scheme 2a). After extensive efforts, we found that DFAS **1a** could be practically obtained on a 10-gram scale with 56 % overall yield through a two-step procedure: reaction of TFP with sodium 4-methylbenzenethiolate at 120 °C in dioxane efficiently provided 3,3-difluoroallyl tolyl sulfide **2a**, which was subsequently methylated by MeOTf under room temperature to afford **1a**. Similarly, using commercially available 2-bromo-3,3,3-trifluoropropene led to **1b** in 73 % overall yield (two steps) on a 15-gram scale with vinyl bromide intact. The presence of vinyl bromide moiety also offers an attractive opportunity to expand the chemical space (Scheme 2a). Remarkably, these two reagents are white solids and are stable in the air for months. Thus, these advantages should render these reagents practical and accessible to synthetic and medicinal researchers. The structures of **1a** and **1b** were further confirmed by their X-ray crystal structure analysis (Scheme 2a).^[19]

With the bench-stable DFASs in hand, the investigation of their reactivities in constructing both C(sp²)-CF₂R and C(sp³)-CF₂R bonds was conducted. Considering copper is much cheaper and more sustainable than palladium, we chose copper-catalyzed cross-coupling between DFASs and readily available organozinc reagents for our studies (Scheme 2b). DFAS **1a** and 4-biphenylzinc reagent **4a** were chosen as the model substrates (Scheme 2b). After a survey of a series of reaction parameters (for details, see the



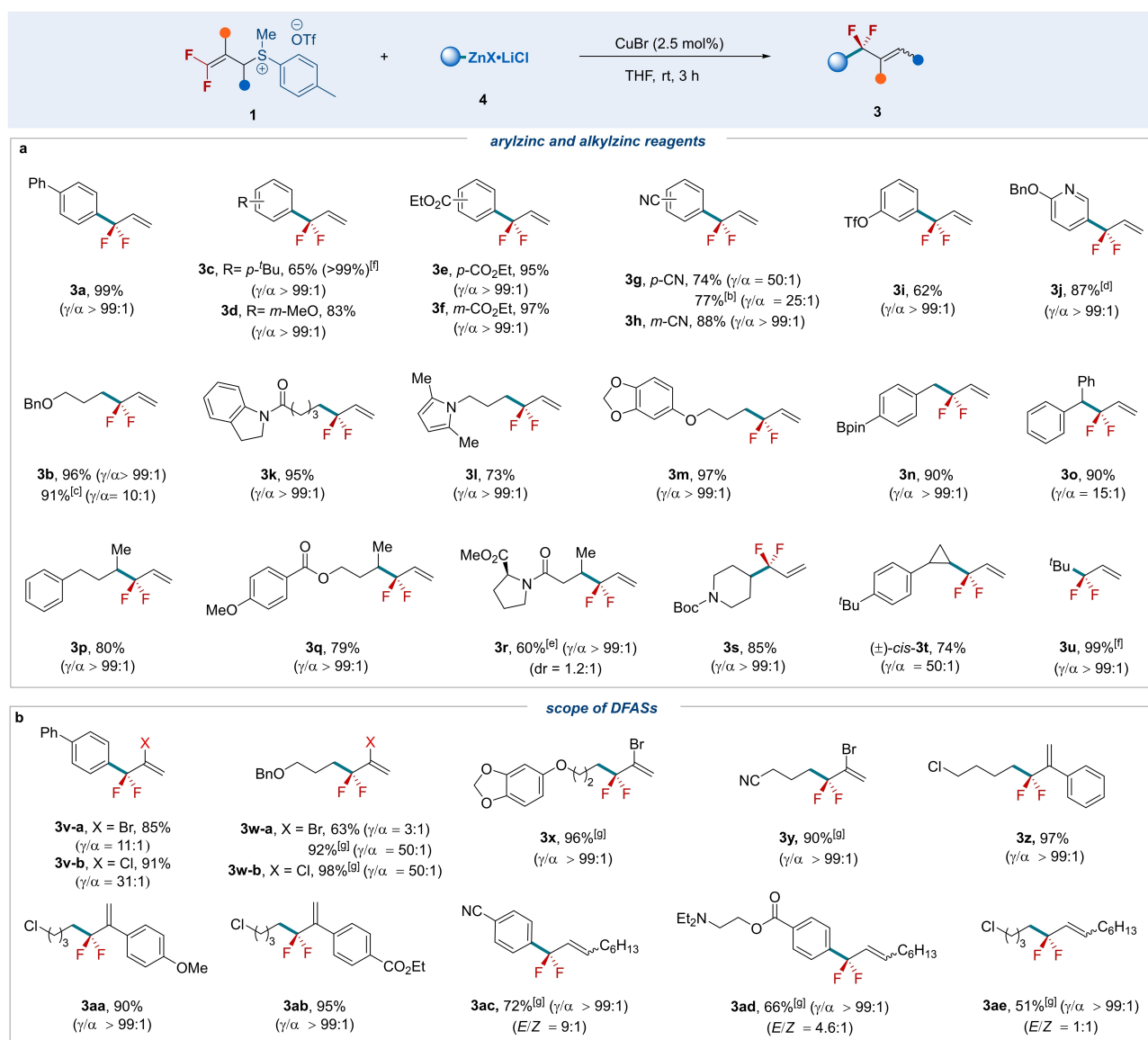
Scheme 2. Preparation of DFASs and copper-catalyzed cross-coupling of organozinc reagents with DFAS **1a** and *gem*-difluoroallyl ammonium salt **1a'**.

Supporting Information), we found that the product **3a** was exclusively provided in almost quantitative yield catalyzed by 2.5 mol% CuBr in THF at room temperature without observation of regioisomer **5a** and methylation side product (Scheme 2b). Decreasing the loading amount of CuBr to 1 mol% led to a comparable yield (97 %) (Scheme 2b). The reaction was not sensitive to the copper sources, as CuX (X = I, Cl) and Cu(OAc)₂ were all applicable to the reaction, with comparable yields obtained (Supporting Information, Table S3). However, the absence of copper salt essentially failed to afford **3a** (Scheme 2b), indicating that a copper-catalyzed process instead of a simple S_N2' pathway is involved in the current reaction. The coupling of **1a** with aryl Grignard reagent **4a'** under standard reaction conditions was also examined (Scheme 2c), providing **3a** in low yield (35 %) with γ/α = 2.2:1. Increasing the CuBr loading from 2.5 mol% to 10 mol% improved the yield of **3a** to 62 % with higher regioselectivity (γ/α = 4.8:1). These findings are in sharp contrast to the previous copper-catalyzed allylation reaction between Grignard reagents and allyl sulphonium salts, wherein α-substitution is a major pathway,^[20] thus featuring the unique properties of fluorinated substrates. Notably, *gem*-difluoroallyl ammonium salt **1a'** showed no activity towards aryl zinc reagent **4a** (Scheme 2c). Although **1a'** has been used as a suitable reagent for S_N2' reaction with heteroatomic nucleophiles,^[14b] the current result demonstrates that DFAS **1a** is more reactive than its corresponding ammonium salts. The cross-coupling of alkyl zinc reagent **4b** with **1a** has also been examined (Scheme 2d). An excellent yield (96 %) of desired product **3b** was obtained without observation of regioisomer

5b. Instead, no **3b** was observed using **1a'** (Scheme 2d), thus demonstrating the synthetic utility of **1a** further.

Substrate Scope of the Copper-Catalyzed Cross-Coupling of Organozinc Reagents with DFASs

Encouraged by the high reactivity of DFAS **1a**, we next examined the substrate scope of this copper-catalyzed coupling (Scheme 3a). Generally, the *gem*-difluoroallylated arenes were obtained in good to excellent yields and with excellent regioselectivity (γ/α > 99:1) (**3a**, **3c–3f**, **3h–3j**). The reaction exhibited good functional group tolerance. Versatile synthetic handles, such as ester, cyano, and triflate, were compatible with the reaction conditions (**3e–3i**). In particular, the survival of triflate (**3i**) demonstrates the synthetic advantage of the current protocol, as triflate is prone to undergo oxidative addition under the previous palladium-catalyzed process.^[15] Pyridine-containing substrate was also applicable to the reaction, with a high yield obtained (**3j**). In addition to aryl zinc substrates, a variety of primary alkyl zinc reagents were also competent coupling partners, wherein the chain length did not interfere with the reaction efficiency (**3b**, **3k–3n**). Indoline, pyrrole, benzo-dioxole, and boronate-derived primary alkyl zinc reagents underwent the *gem*-difluoroallylation efficiently with excellent regioselectivity (**3k–3n**). Secondary alkyl zinc reagents were also amenable to the reaction (**3o–3t**). Sterically hindered substrates, such as phenyl substituted benzylzinc reagent, furnished product **3o** with high efficiency (90 % yield, γ/α = 15:1). Even amino acid-containing secondary alkyl zinc reagent provided desired product **3r** in good yield

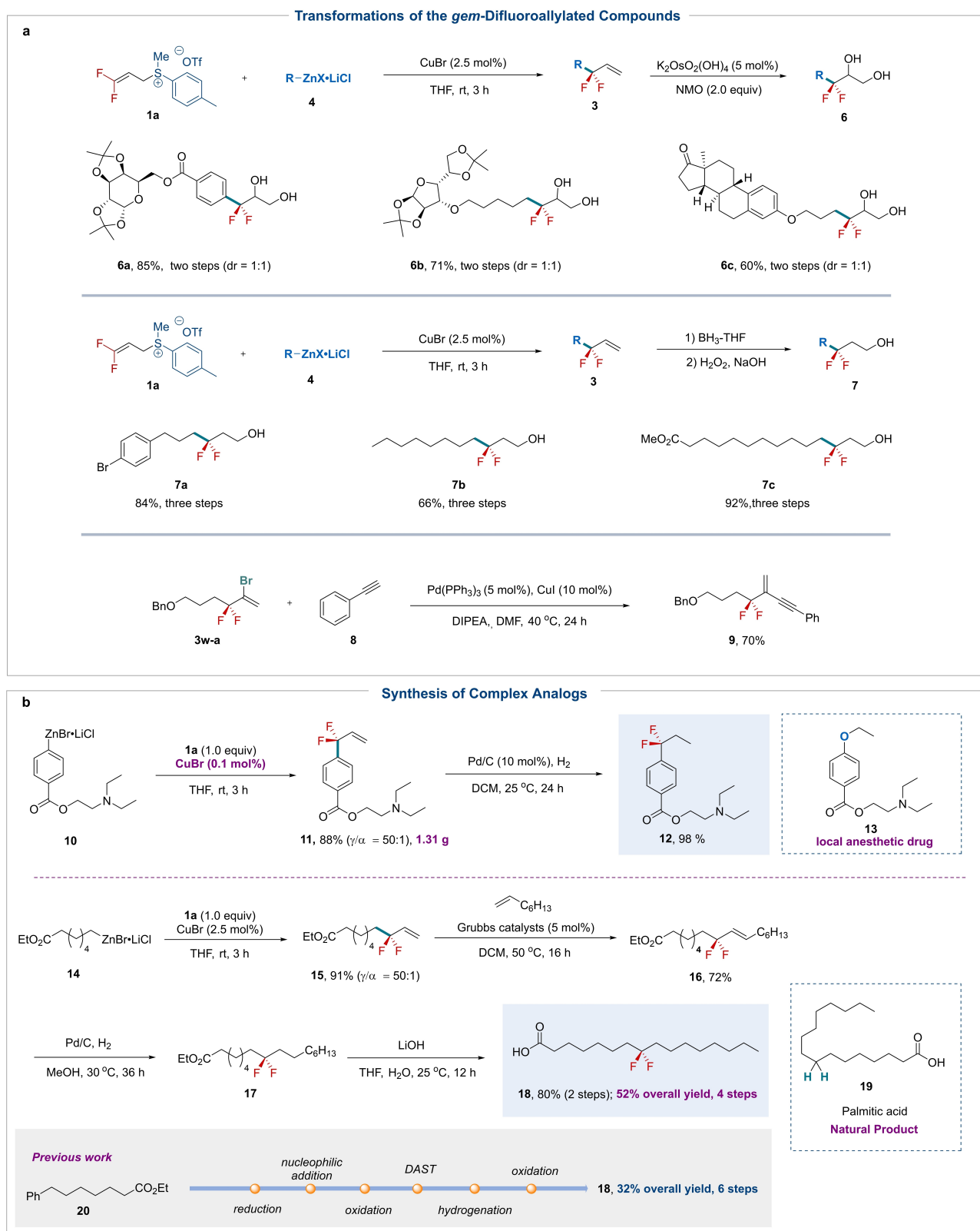


Scheme 3. Copper-catalyzed cross-coupling of organozinc reagents with DFASs.^[a] [a] Reaction conditions (unless otherwise specified): **1** (0.3–0.5 mmol, 1.0 equiv), **4** (1.2 equiv), THF (2.0 mL), 3 h. All reported yields are isolated yields. [b] The reaction was conducted on a 6 mmol scale using 0.1 mol% CuBr. [c] The reaction was conducted on a 4.2 mmol scale using 0.5 mol% CuBr. [d] 1.0 equiv of heteroarylzinc reagent (0.5 mmol) and 1.5 equiv of **1a** were used. [e] 10 mol% CuBr was used, and the reaction was stirred for 12 h. [f] The yield was determined by ¹⁹F NMR. [g] 0.6 equiv of CuBr was used.

and excellent regioselectivity. The reaction was not restricted to acyclic substrates, as cyclic alkyl zinc reagents were also applicable to the reaction (**3s**, **3t**). Piperidine-derived zinc reagent afforded compound **3s** in 85% yield with $\gamma/\alpha > 99:1$. *cis*-Cyclopropylzinc reagent^[21] was also a suitable substrate, providing *cis*-**3t** efficiently. Since cyclopropane moiety exists in numerous natural products and biologically active molecules,^[22] and fluoroalkylated cyclopropanes have been used in the production of pharmaceuticals, such as anti-HCV reagent Glecaprevir^[8] (Scheme 1a), the current protocol provides an efficient route for applications in medicinal chemistry. Remarkably, the alkyl zinc reagents can also be extended to tertiary substrates, as demonstrated by the formation of **3u** in almost quantitative

yield from the reaction of **1a** with *tert*-butyl zinc reagent. The reaction was also scalable. For instance, the 6 mmol-scale synthesis of **3g** provided a comparable yield (77%) with high regioselectivity. It is noteworthy that only 0.1 mol% CuBr was used in this reaction. Although efforts have been made in the copper-catalyzed fluoroalkylation reactions,^[3] the use of low copper catalyst loading remains challenging. To the best of our knowledge, no example of a copper-catalyzed fluoroalkylation reaction using 0.1 mol% copper catalyst has been reported so far. Similarly, the 4 mmol-scale synthesis of **3b** with 0.5 mol% CuBr also proceeded smoothly.

To further demonstrate the generality of this protocol, we examined different DFASs (Scheme 3b). The reaction of



Scheme 4. Transformations of the *gem*-difluoroallylated compounds and synthesis of complex analogs.

vinyl bromide-containing DFAS **1b** with organozinc reagents proceeded efficiently (**3v–3y**). A high yield (85 %) of **3w–a** was obtained by reaction of aryl zinc reagent with **1b**.

But the lower yield (63 %) and regioselectivity ($\gamma/\alpha=3:1$) of **3w–a** were obtained when **1b** was treated with alkyl zinc reagent under standard reaction conditions, suggesting the

different reactivity of **1b** compared with **1a**. Increasing the loading amount of CuBr to 0.6 equiv could increase the yield of **3w-a** to 92 % with excellent regioselectivity ($\gamma/\alpha=50:1$). Similarly, **3x** and **3y** were also obtained efficiently. Higher yields and regioselectivities (**3v-b** and **3w-b**) were obtained using vinyl chloride-containing DFAS **1c**, indicating its higher reactivity. In addition, the α -aryl substituted DFASs bearing electron-donating or electron-withdrawing substituents were also applicable to the coupling, providing the corresponding products **3z-3ab** with high efficiency ($\gamma/\alpha>99:1$). Finally, the investigation of the reactivity of alkyl substituted linear DFASs was conducted, providing **3ac-3ae** in good yields as a Z/E mixture. Notably, these reagents were also applicable to the palladium-catalyzed Suzuki–Miyaura reaction with excellent regioselectivity ($\gamma/\alpha>99:1$), as exemplified by the reaction of aryl boronic acid with **1a** (see the Supporting Information).

Transformations of the *gem*-Difluoroallylated Compounds and Synthesis of Complex Analogs

The synthetic utility of this approach has also been demonstrated by the transformations of the resulting *gem*-difluoroallylated compounds. As shown in Scheme 4a, the reaction of carbohydrate or estrone-derived organozinc reagents with **1a** provided the corresponding *gem*-difluoroallylated products efficiently under standard reaction conditions, which subsequently underwent dihydroxylation to provide diols **6a-6c** in high yields. We could also prepare a series of alcohols **7a-7c** with high efficiency from the reaction of **1a** with alkyl zinc reagents, followed by borylation and oxidation (Scheme 4a). Since fluoroalkylated alcohols are versatile linchpins, these transformations should find applications in the synthesis of complex fluorinated molecules of medicinal interest. In addition, the vinyl bromide-containing *gem*-difluoroallylated compounds can also serve as useful building blocks for diversified synthesis. For instance, Sonagarshia reaction of **3w-a** with alkyne **8** provided enyne **9** efficiently (Scheme 4a), a versatile synthon that otherwise is difficult to prepare by conventional methods. This transformation features the synthetic versatility of DFAS **1b**, as it provides alternative access to a wide range of different *gem*-difluoroallylated compounds by simple transformations of vinyl bromide without the synthesis of different DFASs.

The promising functional compatibility and synthetic efficiency of the current protocol also encouraged us to evaluate its applications in the rapid modification of biologically active molecules and natural products. Given that CF₂ is a bioisostere of an oxygen atom,^[1c,6] we prepared an analog of local anesthetic drug **13**^[23] by replacing its phenoxy oxygen atom with CF₂ in only two steps (Scheme 4b), providing a potential opportunity to discover new bioactive molecules. Remarkably, in this process, a gram-scale synthesis of key intermediate **11** was conducted with high efficiency and excellent regioselectivity ($\gamma/\alpha=50:1$) using CuBr loading as low as 0.1 mol%, thus providing a practical approach to synthetic and medicinal chemistry. We

also prepared a CF₂-modified natural product palmitic acid **18** (Scheme 4b), previously used for the studies of CF₂-effect on molecular conformation changes.^[24] Compared with the previous work that needed 6 steps with 32 % overall yield to prepare **18**,^[24] the current route used copper-catalyzed *gem*-difluoroallylation as the key step, providing **18** in four steps with 52 % overall yield, thus demonstrating the synthetic advantage of the current process. In addition, the successful catalytic olefin metathesis of the *gem*-difluoroallylated compound **15** with alkene in this synthetic route also provides a complementary method to the linear *gem*-difluoroallylated compounds without the preparation of alkyl substituted DFASs.

Conclusion

In conclusion, we have developed a new type of practical and bench-stable *gem*-difluoroallylating reagent DFAS, which can be practically prepared from TFP on a large scale and shows higher reactivity than its corresponding ammonium salts. The reaction of DFAS with aromatic and aliphatic zinc reagents via copper catalysis provides general and practical access to a wide range of *gem*-difluoroallylated compounds. This reaction can also be extended to a series of substituted DFASs, allowing the synthesis of various *gem*-difluoroallylated structures that can serve as versatile building blocks for diversified transformations. The approach overcomes the previous limitations in the construction of C(sp³)-CF₂R bonds and possesses the advantages of synthetic convenience, excellent regioselectivity, good functional group tolerance, and broad substrate scope, even towards tertiary alkyl zinc reagents. In particular, the gram-scale synthesis of the *gem*-difluoroallylated compound with the copper catalyst loading as low as 0.1 mol % features the practicability of this approach. The diversified transformations of the resulting *gem*-difluoroallylated compounds and short synthesis of analogs of the biologically active molecule and natural product further demonstrate the synthetic utility of DFASs. This new type of fluoroalkylating reagent DFASs and the copper-catalyzed coupling are anticipated to have practical applications in the synthesis of complex fluoroalkylated molecules of great interest in life and materials sciences.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Copper Catalysis · Cross-Coupling · *gem*-Difluoroallylation · Organozinc Reagents · Sulfonium Salts

- [1] For selected reviews, see: a) D. O'Hagan, *Chem. Soc. Rev.* **2008**, 37, 308–319; b) W. K. Hagmann, *J. Med. Chem.* **2008**, 51, 4359–4369; c) N. A. Meanwell, *J. Med. Chem.* **2018**, 61, 5822–5880.
- [2] For selected reviews, see: a) J. Wang, M. Sánchez-Roselló, J. L. Aceña, C. del Pozo, A. E. Sorochinsky, S. Fustero, V. A. Soloshonok, H. Liu, *Chem. Rev.* **2014**, 114, 2432–2506; b) M. Inoue, Y. Sumii, N. Shibata, *ACS Omega* **2020**, 5, 10633–10640; c) Y. Ogawa, E. Tokunaga, O. Kobayashi, K. Hirai, N. Shibata, *iScience* **2020**, 23, 101467; d) R. Berger, G. Resnati, P. Metrangola, E. Weber, *Hulliger, Chem. Soc. Rev.* **2011**, 40, 3496–3508.
- [3] For selected reviews, see: a) T. Furuya, A. S. Kamlet, T. Ritter, *Nature* **2011**, 473, 470–477; b) O. A. Tomashenko, V. V. Grushin, *Chem. Rev.* **2011**, 111, 4475–4521; c) C. Alonso, E. M. Marigorta, G. Rubiales, F. Palacios, *Chem. Rev.* **2015**, 115, 1847–1935; d) X. Yang, T. Wu, R. J. Phipps, F. D. Toste, *Chem. Rev.* **2015**, 115, 826–870; e) C. Ni, M. Hu, J. Hu, *Chem. Rev.* **2015**, 115, 765–825.
- [4] For selected reviews, see: a) G. S. Lal, G. P. Pez, R. G. Syvret, *Chem. Rev.* **1996**, 96, 1737–1756; b) J. Charpentier, N. Fruh, A. Togni, *Chem. Rev.* **2015**, 115, 650–682; c) G. K. S. Prakash, M. Mandal, *J. Fluorine Chem.* **2001**, 112, 123–131; d) M. Li, X.-S. Xue, J.-P. Cheng, *Acc. Chem. Res.* **2020**, 53, 182–197.
- [5] a) Z. Fang, Y.-L. Xiao, X. Zhang, *Acc. Chem. Res.* **2018**, 51, 2264–2278; b) B. Chen, D. Vicic, *Top. Organomet. Chem.* **2014**, 52, 113–141.
- [6] a) D. O'Hagan, Y. Wang, M. Skibinski, A. M. Z. Slawin, *Pure Appl. Chem.* **2012**, 84, 1587–1595; b) G. M. Blackburn, D. A. England, F. Kolkman, *J. Chem. Soc. Chem. Commun.* **1981**, 930–932; c) G. M. Blackburn, D. E. Kent, F. Kolkman, *J. Chem. Soc. Perkin Trans. 1* **1984**, 1119–1125.
- [7] a) R. Mondal, M. Agbaria, Z. Nairoukh, *Chem. Eur. J.* **2021**, 27, 7193–7213; b) S. Holovach, K. P. Melnykov, A. Skreminskiy, M. Herasymchuk, O. Tavlii, D. Alosyn, P. Borysko, A. B. Rozhenko, S. V. Ryabukhin, D. M. Volochnyuk, O. O. Grygorenko, *Chem. Eur. J.* **2022**, 28, e202200331.
- [8] a) J. O. Link, J. G. Taylor, L. Xu, M. Mitchell, H. Guo, H. Liu, D. Kato, T. Kirschberg, J. Sun, N. Squires, J. Parrish, T. Kellar, Z. Y. Yang, C. Yang, M. Matles, Y. Wang, K. Wang, G. Cheng, Y. Tian, E. Mogalian, E. Mondou, M. Cornpropst, J. Perry, M. C. Desai, *J. Med. Chem.* **2014**, 57, 2033–2046; b) Y. N. Lamb, *Drugs* **2017**, 77, 1797–1804.
- [9] Y. Matsumura, N. Mori, T. Nakano, H. Sasakura, T. Matsugi, H. Hara, Y. Morizawa, *Tetrahedron Lett.* **2004**, 45, 1527–1529.
- [10] a) P. Tarrant, M. Lovelace, *J. Am. Chem. Soc.* **1954**, 76, 3466–3468; b) D. Seyferth, R. M. Simon, D. J. Sepelak, H. A. Klein, *J. Org. Chem.* **1980**, 45, 2273–2274.
- [11] a) Z. Yang, D. J. Burton, *J. Org. Chem.* **1991**, 56, 1037–1041; b) X. Zhang, H. Xia, X. Dong, J. Jin, W.-D. Meng, F.-L. Qing, *J. Org. Chem.* **2003**, 68, 9026–9033; c) X. Yue, X. Zhang, F.-L. Qing, *Org. Lett.* **2009**, 11, 73–76.
- [12] Q.-Q. Min, Z. Yin, Z. Feng, W.-H. Guo, X. Zhang, *J. Am. Chem. Soc.* **2014**, 136, 1230–1233.
- [13] a) M. Kirihara, T. Takuwa, M. Okumura, T. Wakikawa, H. Takahata, T. Momose, Y. Takeuchi, H. Nemoto, *Chem. Pharm. Bull.* **2000**, 48, 885–888; b) F. Tellier, M. Baudry, R. Sauvetre, *Tetrahedron Lett.* **1997**, 38, 5989–5992.
- [14] a) L. Tang, Z.-Y. Liu, W. She, C. Feng, *Chem. Sci.* **2019**, 10, 8701–8705; b) F. Ye, Y. Ge, A. Spannenberg, H. Neumann, L.-W. Xu, M. Beller, *Nat. Commun.* **2021**, 12, 3257.
- [15] a) Y.-G. Lou, A.-J. Wang, L. Zhao, L.-F. He, X.-F. Li, C.-Y. He, X. Zhang, *Chem. Commun.* **2019**, 55, 3705–3708; b) S. Sakamoto, T. W. Butcher, J. L. Yang, J. F. Hartwig, *Angew. Chem. Int. Ed.* **2021**, 60, 25746–25752; *Angew. Chem.* **2021**, 133, 25950–25956.
- [16] a) X. Ren, X. Gao, Q.-Q. Min, S. Zhang, X. Zhang, *Chem. Sci.* **2022**, 13, 3454–3460; b) W.-J. Yue, C. S. Day, A. J. B. Rucinski, R. Martin, *Org. Lett.* **2022**, 24, 5109–5114.
- [17] D. Kaiser, I. Klose, R. Oost, J. Neuhaus, N. Maulide, *Chem. Rev.* **2019**, 119, 8701–8780.
- [18] For selected examples, see: a) T. Umemoto, *Chem. Rev.* **1996**, 96, 1757–1778; b) J. J. Yang, R. L. Kirchmeier, J. M. Shreeve, *J. Org. Chem.* **1998**, 63, 2656–2660; c) A. Matsnev, S. Noritake, Y. Nomura, E. Tokunaga, S. Nakamura, N. Shibata, *Angew. Chem. Int. Ed.* **2010**, 49, 572–576; *Angew. Chem.* **2010**, 122, 582–586; d) G. K. S. Prakash, C. Weber, S. Chacko, G. A. Olah, *Org. Lett.* **2007**, 9, 1863–1866; e) Y. Liu, X. Shao, P. Zhang, L. Lu, Q. Shen, *Org. Lett.* **2015**, 17, 2752–2755; f) S.-L. Lu, X. Li, W.-B. Qin, J.-J. Liu, Y.-Y. Huang, H. N. C. Wong, G.-K. Liu, *Org. Lett.* **2018**, 20, 6925–6929.
- [19] Deposition numbers 2183276 (**1a**) and 2183277 (**1b**) contain the supplementary crystallographic data for this paper. These data are provided free of charge by the joint Cambridge Crystallographic Data Centre and Fachinformationszentrum Karlsruhe Access Structures service.
- [20] Y. Gendreau, J. F. Normant, J. Villieras, *J. Organomet. Chem.* **1977**, 142, 1–7.
- [21] L. An, F.-F. Tong, S. Zhang, X. Zhang, *J. Am. Chem. Soc.* **2020**, 142, 11884–11892.
- [22] a) T. T. Talele, *J. Med. Chem.* **2016**, 59, 8712–8756; b) A. Gagnon, M. Duplessis, L. Fader, *Org. Prep. Proced. Int.* **2010**, 42, 1–69.
- [23] D. Chen, K. P. Taylor, Q. Hall, J. M. Kaplan, *Genetics* **2016**, 204, 1151–1159.
- [24] Y. Wang, R. Callejo, A. M. Z. Slawin, D. O'Hagan, *Beilstein J. Org. Chem.* **2014**, 10, 18–25.

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